

GEOSPATIAL WEBSERVICES

HOW TO FIND WHAT YOU'RE LOOKING FOR



SLIDES

<https://slides.michelstuyts.be>

INTRODUCTION

MICHEL STUYTS



INTRODUCTION

MICHEL STUYTS



- 25 years of GIS experience
- 14 years of QGIS experience
- 12 years as GIS coordinator at local municipalities
- Since 1 january 2024 working for Geo-IT bv in Antwerp

WHERE TO FIND ME?

- mail: info@michelstuyts.be
- website: <https://michelstuyts.be>
- blog: <https://stuyts.xyz>
- mastodon: <https://fosstodon.org/@michelstuyts>
- linkedin: <https://www.linkedin.com/in/michelstuyts/>

GEO-IT BV

GIS

GEO-IT BV

GIS

- GIS software based on:
 - MapGuide Open Source
 - OpenLayers
 - PostGIS

GEO-IT BV

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 - PostGIS
- GIS workshops

GEO-IT BV

GIS

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 - PostGIS
- GIS workshops
- GIS support

GEO-IT BV

CAD

- Reseller of Autodesk software
- AutoCAD Add-ons
- Courses on Autodesk software

SUBJECT OF THIS PRESENTATION

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- A personal side project

SUBJECT OF THIS PRESENTATION

- A personal side project
- Information is available online under the CC0 1.0 Universal License

WHAT?

- An online collection of information about geospatial webservices with data for Belgium
- Started 10 years ago

WHAT?

Seperate websites for:

- WMS
- WFS
- WMTS
- OGC API Features services
- xyz Tile Layers

WHY?

WHY?

- I wanted to have a single source for all relevant geospatial webservices

WHY?

- I wanted to have a single source for all relevant geospatial webservices
- I wanted to share this information with others

HOW DID I START?

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- Started with an offline spreadsheet

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- Moved to a an online MySQL database

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- Made the first simple website in PHP with lots of iframes

HOW DID I START?

- Started with an offline spreadsheet
- Moved to a an online MySQL database
- Made the first simple website in PHP with lots of iframes
- I created seperate websites on different subdomains per type of webservice

HOW DO I COLLECT THESE WEBSERVICES?

10 YEARS AGO

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HOW DO I COLLECT THESE WEBSERVICES?

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- I collected all the url's from official geospatial data providers
- I looked in the source of web maps

HOW DO I COLLECT THESE WEBSERVICES?

TODAY

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- I created scripts to search the API's of official geospatial agencies
- I use a crawler to search on ArcGIS servers for WMS, WMTS and WFS services based on Scrapy

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- I still look at the source code of web maps

HOW DO I COLLECT THESE WEBSERVICES?

TODAY

- I created scripts to search the API's of official geospatial agencies
- I use a crawler to search on ArcGIS servers for WMS, WMTS and WFS services based on Scrapy
- I still look at the source code of web maps
- People send me new urls they found or created

HOW DO I MAINTAIN THE DATA?

10 YEARS AGO

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HOW DO I MAINTAIN THE DATA?

10 YEARS AGO

- I created simple PHP scripts to check the availability of web services using a cron job
- I also checked the speed of the services by requesting the GetCapabilities
- I get the title, description and all other details from the GetCapabilities response

HOW DO I MAINTAIN THE DATA?

TODAY

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HOW DO I MAINTAIN THE DATA?

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- I check for the availability of https versions
- I manually remove services that are offline
- I no longer show the speed of the services

HOW DO I MAINTAIN THE DATA?

TODAY

- I optimized the PHP scripts for better performance
- I check for changes in title, description, ...
- I check for the availability of https versions
- I manually remove services that are offline
- I no longer show the speed of the services
- Only first time the title from the GetCapabilities, after that manually

HOW DO I MAINTAIN THE DATA?

TODAY

WMS Service Status Check

Checking WMS services for availability and capabilities...

Service ID: 1455

<https://data.geo.be/ws/rbins/wms>

Capabilities URL: <https://data.geo.be/ws/rbins/wms?request=GetCapabilities&service=WMS>

Status: **OK**

Laag1: benthichabitatconditiondatasetinsupportofthemarinestrategyframeworkdirective_msfd_descriptor6_d06c05_2018reporting_

Responsetijd: 0.064581

Grootte: 40256

Snelheid: 608.73 kB/s

HTTP Code: 200

Service ID: 1310

https://geoservices.wallonie.be/arcgis/services/INDUSTRIES_SERVICES/ZONES_DEVELOPPEMENT/MapServer/WMSServer

Capabilities URL: https://geoservices.wallonie.be/arcgis/services/INDUSTRIES_SERVICES/ZONES_DEVELOPPEMENT/MapServer/WMSServer?request=GetCapabilities&service=WMS

Status: **OK**

Laag1: 0

Responsetijd: 1.67833

Grootte: 10588

Snelheid: 6.16 kB/s

HTTP Code: 200

HOW DO I MAINTAIN THE DATA?

FREQUENCY OF CHECKS

- WMS: 2 services every 30 minutes
- WMTS: 2 services every 30 minutes
- WFS: 3 services every 30 minutes
- OGC API Features: 2 services every 60 minutes
- xyz Tile Layers: 27 zoom levels for 1 service every 30 minutes

WHERE TO FIND THE INFO?

[HTTPS://WEBSERVICES.MICHELSTUYTS.BE](https://webservices.michelstuyts.be)

- <https://wms.michelstuyts.be>
- <https://wfs.michelstuyts.be>
- <https://features.michelstuyts.be>
- <https://wmts.michelstuyts.be>
- <https://xyz.michelstuyts.be>

Multilingual: English, Dutch, French

WHERE TO FIND THE INFO?



Geographical Web Services

About this website

This page gives a list of websites that have an overview of geographical services with data for Belgium. For more information click on one of the types below. These overviews are compiled by [Michel Stuyts](#).

Types of Web Services

- [WMS](#)
- [WFS](#)
- [WMTS](#)
- [OGC API Features services](#)
- [xyz Tile Layers](#)

webservices.michelstuyts.be

STATISTICS

- WMS: 838 services
- WFS: 261 services
- WMTS: 147 services
- OGC API Features services: 166 services
- xyz Tile Layers: 200 services
- Total: 1612 webservices

A LOOK AT THE WEBSITES

A LOOK AT THE WEBSITES

- All websites have lists of webservices per region

A LOOK AT THE WEBSITES

- All websites have lists of webservices per region
 - Belgium
 - Flanders
 - Wallonia
 - Brussels
 - Worldwide

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- All websites have QGIS import files

A LOOK AT THE WEBSITES

- All websites have lists of webservices per region
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 - Worldwide
- All websites have search functionality
- All websites have QGIS import files
- All data also available on Codeberg

LISTS


[QGIS import file](#)
[Data on Codeberg](#)
[Other services](#)

[NL](#) - [FR](#) - [EN](#)



WMS Services with data for Belgium

This page gives an overview of [WMS - Web Map Services](#) with data for Belgium. For more information click on the name of one of the services below. If you know a WMS service that is missing from this list, please send it to [Michel Stuyts](#).

The data from this website is also available on [Codeberg](#).

The data on this website can also be downloaded for use in [QGIS](#) (version 3.2 or newer). This [XML file](#) with all services can be imported into QGIS. (XML update: 09/09/2025)

These kind of overviews are available for [Web Map Services](#) (WMS) with data for Belgium, [Web Map Tile Services](#) (WMTS) with data for Belgium, [Web Features Services](#) (WFS) with data for Belgium, [OGC API Feature Services](#) with data for Belgium and [xyz Tile Layer Services](#) with data for Belgium.

Services Belgium

[Aeronautical Obstacle Evaluation Map for Belgium \(wms.ngi.be\)](#)

[Cadastral Layers Belgium \(ccff02.minfin.fgov.be\)](#)

[Cadastral parcels Belgium \(ccff02.minfin.fgov.be\)](#)

[CartoWeb.be NGI-IGN \(cartoweb.wms.ngi.be\)](#)

[Digital surface model Meuse, Vesdre and Demer \(wms.ngi.be\)](#)

[eTOD database \(wms.ngi.be\)](#)

[Flanders Marine Institute \(VLIZ\) - Geoserver WMS Service \(geo.vliz.be\)](#)

LISTS

 QGIS import file Data on Codeberg Other services Search a service Search NL - FR - EN



WMS Services with data for Belgium

Mercator Publieke View Service

Publieke View Service van MercatorNet

url:

<https://mercator.vlaanderen.be/raadpleegdienstenmercatorpubliek/us/wms>

GetCapabilities

<https://mercator.vlaanderen.be/raadpleegdienstenmercatorpubliek/us/wms?REQUEST=GetCapabilities&SERVICE=WMS>

Region:

Flanders

Maintainer:

Vlaamse Overheid - Beleidsdomein Omgeving - MercatorNet

Available layers:

SEARCH FUNCTIONALITY


[QGIS import file](#)
[Data on Codeberg](#)
[Other services](#)

[NL](#) - [FR](#) - [EN](#)



WFS Services with data for Belgium

Type of Search:

☐ Exact sentence
☒ All words
☐ Any of the words

Search results

[BE-LB WFS service - wfs.environnement.brussels \(Brussels\)](#)
[Bruenvi / Service de visualisation WFS - ows.environnement.brussels \(Brussels\)](#)
[geoservices-urbis.irisnet.be - geoservices-urbis.irisnet.be \(Brussels\)](#)
[Vervoersnetwerken weg - geo.api.vlaanderen.be \(Flanders\)](#)
[GeoServer Web Feature Service - geo.onroerenderfgoed.be \(Flanders\)](#)
[GeoServer Web Feature Service - geo.onroerenderfgoed.be \(Flanders\)](#)
[Bruit // Geluid - ows.environnement.brussels \(Brussels\)](#)

QGIS IMPORT FILES

```

qgis_wmts.xml
1 <!DOCTYPE connections>
2 <qgsWMSConnections version="1.0">
3   <wms ignoreGetMapURI="false" smoothPixmapTransform="false" dpiMode="7" password=""
    ignoreGetFeatureInfoURI="false" referer="" username="" url=
    "https://cartoweb.wmts.ngi.be/1.0.0/WMTSCapabilities.xml?REQUEST=GetCapabilities&SERV
    ICE=WMTS&version=1.0.0" invertAxisOrientation="false" ignoreAxisOrientation="false"
    name="Bel - CartoWeb.be (NGI - IGN) (WMTS)"/>
4   <wms ignoreGetMapURI="false" smoothPixmapTransform="false" dpiMode="7" password=""
    ignoreGetFeatureInfoURI="false" referer="" username="" url=
    "https://geo.bipt-data.be/geoserver/gwc/service/wmts?REQUEST=GetCapabilities&SERVICE=
    WMTS&version=1.0.0" invertAxisOrientation="false" ignoreAxisOrientation="false" name=
    "Bel - GeoServer Web Map Tile Service (GeoServer) (WMTS)"/>
5   <wms ignoreGetMapURI="false" smoothPixmapTransform="false" dpiMode="7" password=""
    ignoreGetFeatureInfoURI="false" referer="" username="" url=
    "https://www.verkeerscentrum.be/geo/gwc/service/wmts?REQUEST=GetCapabilities&SERVICE=
    WMTS&version=1.0.0" invertAxisOrientation="false" ignoreAxisOrientation="false" name=
    "Bel - GeoServer Web Map Tile Service (www.verkeerscentrum.be) (WMTS)"/>
6   <wms ignoreGetMapURI="false" smoothPixmapTransform="false" dpiMode="7" password=""
    ignoreGetFeatureInfoURI="false" referer="" username="" url=
    "https://wmts.ngi.be/inspire/ortho/1.0.0/WMTSCapabilities.xml?REQUEST=GetCapabilities&
    ;SERVICE=WMTS&version=1.0.0" invertAxisOrientation="false" ignoreAxisOrientation=
    "false" name="Bel - View service - Orthophotos (NGI - IGN) (WMTS)"/>
7   <wms ignoreGetMapURI="false" smoothPixmapTransform="false" dpiMode="7" password=""
    ignoreGetFeatureInfoURI="false" referer="" username="" url=
    "https://tiles.arcgis.com/tiles/wD478ElH9JFhKjT3/arcgis/rest/services/Brussels_3D_WebScen
    e_WTL1/MapServer/WMTS/1.0.0/WMTSCapabilities.xml?REQUEST=GetCapabilities&SERVICE=WMTS
    &version=1.0.0" invertAxisOrientation="false" ignoreAxisOrientation="false" name="Br
    - Brussels_3D_WebScene_WTL1 (tiles.arcgis.com) (WMTS)"/>
8   <wms ignoreGetMapURI="false" smoothPixmapTransform="false" dpiMode="7" password=""

```

CODEBERG

The screenshot shows the Codeberg web interface for the repository 'gis-projects / Belgium-XYZ-tiles'. The top navigation bar includes links for Explore, About, FAQ, Help, and a Donate button. On the right, there are links for Register and Sign in. The repository header shows the name 'gis-projects / Belgium-XYZ-tiles' and statistics: 1 Watch, 0 Stars, and 0 Forks. Below the header, there are tabs for Code, Issues, Pull requests, and Activity. The main content area displays a list of xyz Tile Layer Services with data for Belgium, with a link to <https://xyz.michelstuyts.be>. It also shows repository statistics: 85 commits, 1 branch, 0 tags, and a size of 358 KiB. A file browser section shows the 'master' branch with a 'Find a file' button and a list of files: LICENSE (initial commit, 7 years ago) and README.md (Edit README.md, 8 months ago). The README.md content is displayed below, titled 'xyz Tile Layer Services List Belgium', and includes a description of the repository's purpose and a link to the README file.

gis-projects / Belgium-XYZ-tiles

<> Code Issues Pull requests Activity

A list of xyz Tile Layer Services services with data for Belgium. <https://xyz.michelstuyts.be>

85 commits 1 branch 0 tags 358 KiB

master Find a file HTTPS <https://codeberg.org/gis-projects/Belgium-XYZ-tiles.git>

Michel Stuyts 0eb7e81331 Edit README.md 8 months ago

LICENSE [initial commit](#) 7 years ago

README.md Edit README.md 8 months ago

README.md

xyz Tile Layer Services List Belgium

A list of xyz Tile Layer Services with data for Belgium

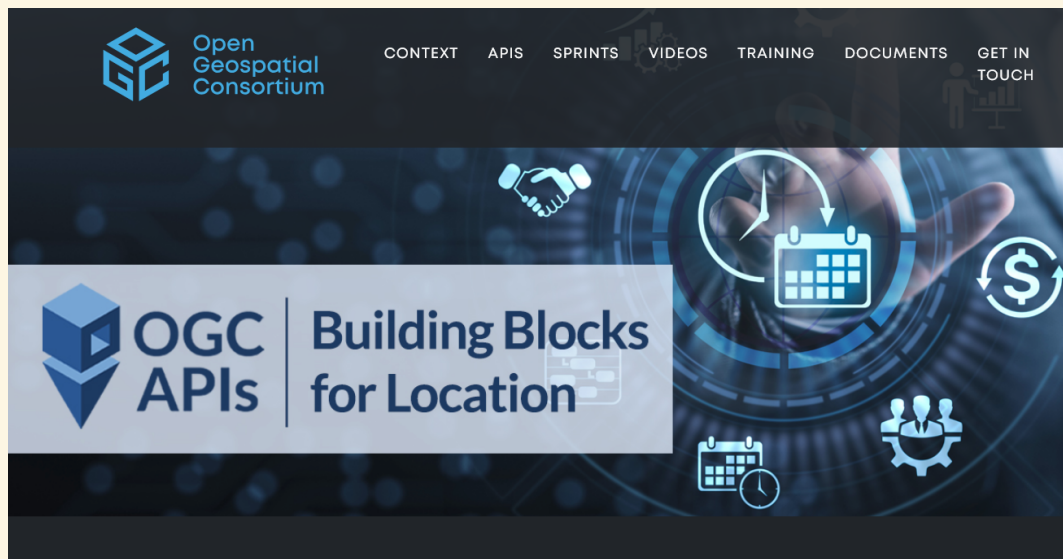
The entire content of this repository is stored in this README.md document. All editing should be done on this file. The only other file is the LICENSE file.

More information about xyz Tile Layer Services has been found on [Wikipedia](#).

<https://codeberg.org/gis-projects/>

THE FUTURE

MAYBE NEW TYPES OF WEBSERVICES

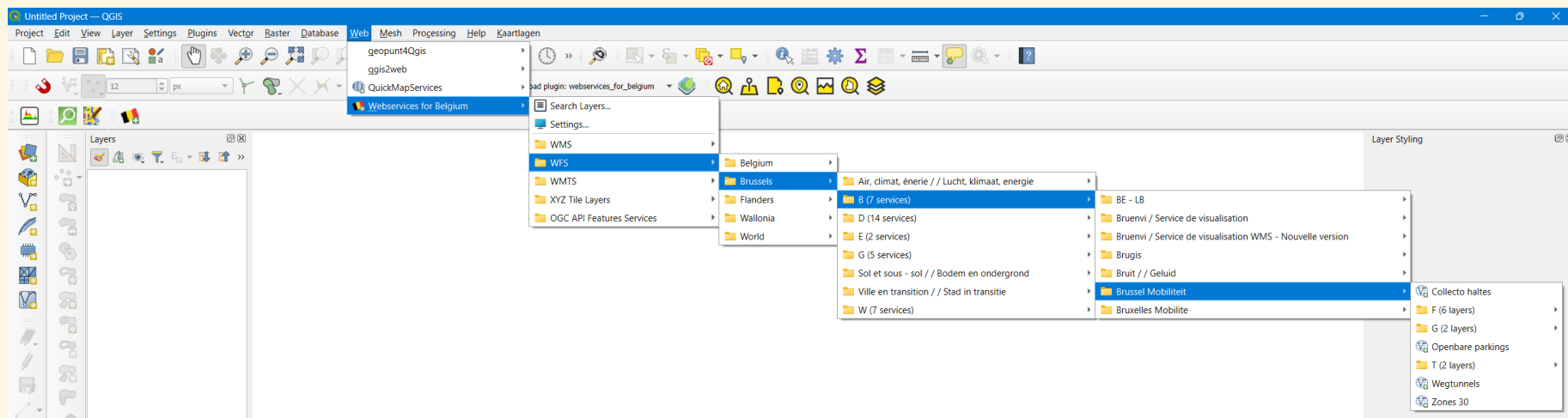


OGC API - MAPS

The OGC API - Maps Standard describes an API that can serve spatially referenced electronic maps, whether static or dynamically rendered, in a manner independent of the underlying data store. OGC API - Maps describes the discovery and query

THE FUTURE

MAYBE A QGIS PLUGIN TO EASILY USE WEBSERVICES



THE FUTURE

MAYBE A QGIS PLUGIN TO EASILY USE WEBSERVICES



Search for layers...



Search:

orthol

Search

Service Types



WMS



WFS



WMTS



XYZ



Features

Search Results (100 found):



Orthophoto Meuse, Vesdre and Demer - Ortho Floods 2021 (WMS) (Belgium)



Orthoimage coverage - 2023 - Orthophoto (WMS) (Belgium)



Orthoimage coverage - 2016 - Orthophoto (WMS) (Belgium)



Orthoimage coverage - 2017 - Orthophoto (WMS) (Belgium)



Orthoimage coverage - 2018 - Orthophoto (WMS) (Belgium)



Orthoimage coverage - 2019 - Orthophoto (WMS) (Belgium)



Orthoimage coverage - 2020 - Orthophoto (WMS) (Belgium)



Orthoimage coverage - 2021 - Orthophoto (WMS) (Belgium)



Orthoimage coverage - 2022 - Orthophoto (WMS) (Belgium)



Orthofotomozaïek, middenschalig, winteropnamen, kleur, 2023, Vlaanderen - Orthobeeldvorming (WMS) (Flanders)



Orthofotomozaïek, middenschalig, zomeropnamen, kleur, 2021, Vlaanderen - Orthobeeldvorming (WMS) (Flanders)



Orthofotomozaïek, middenschalig, zomeropnamen, panchromatisch, 2021, Vlaanderen - Orthobeeldvorming (WMS) (Flanders)



Orthofotomozaïek, middenschalig, zomeropnamen, kleur-infrarood, 2021, Vlaanderen - Orthobeeldvorming (WMS) (Flanders)



ortho_k2006 - Orthofoto archief Oost - Vlaanderen (WMS) (Flanders)



ortho_k2006.jp2 - Orthofoto archief Oost - Vlaanderen (WMS) (Flanders)



ortho_k2009 - Orthofoto archief Oost - Vlaanderen (WMS) (Flanders)



ortho_k2009_5m.jp2 - Orthofoto archief Oost - Vlaanderen (WMS) (Flanders)



ortho_k2012 - Orthofoto archief Oost - Vlaanderen (WMS) (Flanders)

Load All

Load Selected

Close

OTHER PROJECTS

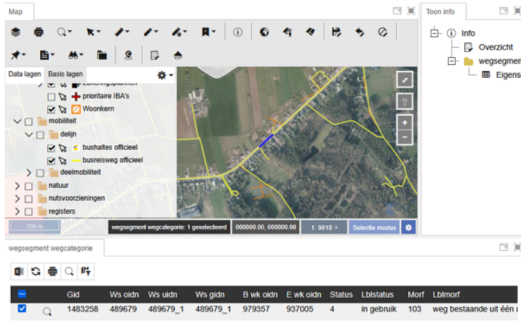
<https://stuyts.xyz>: GIS blog

[Frontpage](#) [About me](#) [English](#) [Nederlands](#)

stuyts.xyz

Michel Stuyts' blog about GIS, geography and geodata





Map

30 years of MapGuide (Open Source)

April 2, 2025 by Michel Stuyts

This year MapGuide is 30 years old. Even though it isn't the most popular geospatial project, it is still worth taking a look at. You might have noticed I put "Open source" between brackets, that's because it started as proprietary software, but later on was made

Change language /
Verander van taal

[English](#)

[Nederlands](#)

[Mastodon](#)

[Pixelfed](#)

[Instagram](#)

[Bluesky](#)

[Codeberg](#)

OTHER PROJECTS

<https://qgis.tips>: QGIS tips and tricks



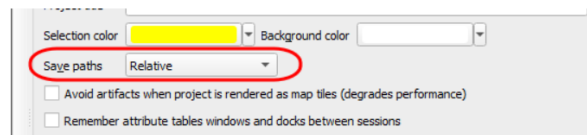
QGIS Tips

Tips & Tricks for QGIS users

[Tips](#)[About](#)[Contact](#)[Mastodon](#)

Absolute vs relative paths in QGIS projects

August 29, 2025 by [Michel Stuyts](#)



The choice between absolute and relative paths is a setting in QGIS projects that I wasn't aware of until the beginning of 2024. At that time I got the course material from a QGIS workshop a co-worker used to teach and which mentioned this option. Since then I have started teaching the QGIS workshops for this company and of course "Absolute and relative paths" is now also part of my workshops.

So is this an important setting? Well, you can use QGIS perfectly well without knowing that this setting exists, but it's good to know about it – as you will see below. This blog post explains what absolute and relative paths are and when you should use each.

[Read more](#)

OTHER PROJECTS

<https://geone.ws>: Geospatial news

GeoNews

GIS, Geospatial and Geodata related news



MapGuide Open Source 4.0 RC1 and MapGuide Portable announced

September 17, 2025



More GeoNe.ws

- GeoNe.ws on Mastodon
- GeoNe.ws on Bluesky
- GeoNe.ws on Blogger
- GeoNe.ws on Tumblr
- GeoNe.ws on Facebook
- Contact GeoNe.ws

Get all the news
in your mailbox

THE END

[HTTPS://WEBSERVICES.MICHELSTUYTS.BE](https://webservices.michelstuyts.be)